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Molecular dynamics insights into the selective extraction of 1,3-propanediol from fermentation broth using eutectic solvent

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ABSTRACT

The selective extraction of 1,3-propanediol (PDO) is vital for improving the sustainability of bio-based chemical production from renewable feedstocks. This study investigates the use of a eutectic solvent (ES) composed of menthol and dodecanoic acid (3:1) for the efficient extraction of PDO from fermentation broth. A total of seven cases were examined, including four experimental and three designed cases, to evaluate co-solute interference and PDO selectivity. Molecular dynamics simulations were employed to assess mass density profiles, hydrogen bonding, interaction energies, diffusivity, and Gibbs free energy of solvation (ΔG). The density profiles confirmed distinct two-phase formation with no solvent loss and minimal water intrusion (~4 wt%) into the ES-rich phase. The ES consistently demonstrated high extraction efficiency for PDO (~80 %) with excellent selectivity, showing selectivity ratios of about 7 compared to glycerol and butanediol, up to 40 compared to organic acids, and over 100 compared to water. Hydrogen bonding and spatial distribution function (SDF) analyses indicated a higher density of PDO around the hydroxyl group of menthol and the carboxylic group of dodecanoic acid, reflecting strong site-specific interactions. Diffusivity calculations showed restricted mobility of PDO in the ES phase, suggesting molecular stabilization through interaction. Interaction energy analysis revealed stronger affinity of PDO for ES, while other co-solutes (BDO, GLY, SUC) preferred water. These findings were further validated by ΔG values, confirming that PDO has a more favorable solvation energy in the ES compared to water, supporting its selective extraction and separation.

1. Introduction

The generation and mismanagement of agricultural waste remain pressing global concerns, particularly in developing countries where biomass residues from fields or agro-industrial processes are commonly left to decompose or are burned in the open. These practices contribute to severe environmental pollution and result in the loss of valuable bioresources. Among the various agricultural residues, sugarcane bagasse (SCB), the fibrous by-product of sugarcane processing, holds significant potential for valorization [1]. With global sugarcane production estimated at approximately 1.6 billion tons annually, around 279 million metric tons of SCB are generated each year [2]. SCB is a rich lignocellulosic biomass composed primarily of cellulose, hemicellulose, and lignin, making it a promising feedstock for producing bio-based chemicals [3]. One such high-value chemical is 1,3-Propanediol (PDO), a platform molecule used to manufacture bioplastics, polymers, cosmetics, and solvents. The global market for PDO was valued at approximately USD 491.97 million in 2023 and is projected to grow at a compound annual growth rate (CAGR) of 7.2 % from 2023 to 2032 [4]. Traditionally, PDO has been synthesized through petrochemical routes, such as those developed by Shell and Degussa [5]. These methods rely on fossil resources and require high temperatures, high pressures, and metal catalysts, leading to several drawbacks, including energy-intensive processes, harsh reaction conditions, and the formation of toxic intermediates.

As sustainability becomes a key driver in the chemical industry, increasing interest is in developing greener production methods for platform chemicals. Recent studies have highlighted the feasibility of producing PDO through the anaerobic fermentation of glycerol, a byproduct of biodiesel production. In particular, the bioconversion of SCB hydrolysate via anaerobic glycerol fermentation presents a circular and eco-friendly approach to agricultural waste utilization [6]. Despite progress in upstream bioconversion, the downstream recovery of PDO from fermentation broth remains a significant challenge. Several

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processes, including evaporation, distillation ([5,6], chromatography, reactive extraction, and even enzymatic methods involving esterification, have been reported for the extraction of PDO, but they often come with certain drawbacks, such as high energy consumption, low selectivity, and the generation of secondary waste [7–12]. Therefore, there is a growing need for efficient and sustainable separation strategies.

Among various industrial separation methods such as crystallization, distillation, and liquid-liquid extraction (LLE), LLE stands out as a more efficient and sustainable technique, offering advantages in selectivity, energy consumption, and environmental impact, thereby driving its growing application in bio-based product recovery. Recent research has highlighted the utilization of ionic liquids (ILs) in the liquid-liquid extraction of industrially important chemicals, demonstrating their potential in various extraction processes [13–16]. Several studies have also shown the high extraction performance of ILs in the recovery of PDO [12,14,17,18], with one of them being phosphonium-based ILs such as ([P66614] [C_8SO_3]), which exhibited a distribution coefficient (β_{PDO}) of 0.219 and a selectivity (S_{PDO}) of 2.65. However, their high costs hinder the practical application of ILs, prompting the exploration of more cost-effective alternatives. A promising alternative to ionic liquids (ILs) is a novel class of solvents known as Eutectic Solvents (ESs). ESs are formed by combining a hydrogen bond acceptor (HBA) and a hydrogen bond donor (HBD), which interact through hydrogen bonding to create a stable liquid with a lower melting point than the individual components [19]. These solvents offer several advantages, including low toxicity, biodegradability, and tunability, making them an attractive, cost-effective, and sustainable option for the selective extraction of PDO. However, many ESs developed so far are choline chloride-based and inherently hydrophilic, which makes them unstable in water-based environments and limits their practical applications. Research has shifted toward developing hydrophobic eutectic solvents to address this issue. Hydrophobic ESs were first introduced in 2015 by Van Osch and colleagues, who combined quaternary ammonium salts with decanoic acid to form water-immiscible solvents [20]. These systems demonstrated improved stability in aqueous media and opened new possibilities for their application in liquid-liquid extraction and other separation processes.

In our earlier study, five hydrophobic ESs were evaluated for their ability to extract PDO from an aqueous system. Among them, the ES, composed of menthol and dodecanoic acid in a 3:1 M ratio (MD (3:1)), exhibited the most promising extraction performance [21–23]. Based on these findings, the present work employs molecular dynamics simulations to investigate the selective extraction of PDO using MD (3:1) from realistic and controlled systems. The realistic system includes PDO and major co-solutes, namely, 2,3-butanediol, glycerol, succinic acid, lactic acid, and citric acid, typically present in the fermentation broth. In contrast, the controlled system maintains equal mass ratios of PDO and co-solutes. The controlled system was used to evaluate the competitive affinity of each component toward the ES. Using molecular simulations, extraction efficiencies of all components were quantified alongside interaction energies, structural and transport properties, and the Gibbs free energy of solvation. These simulations provided detailed molecular-level insights into solute-solvent interactions, elucidating the extraction performance of MD (3:1) under diverse conditions.

2. Computational methodology

2.1. Computational details

Seven different systems were modeled, including four based on experimentally reported fermentative broth compositions (Cases 1 to 4) and three additional systems (Cases 5 to 7) with equal mass ratios of 1,3-propanediol and co-solutes to evaluate competitive affinity. The compositional details of all cases are provided in Table 1, and the specific molecular counts used in each case are listed in Table S1 [21–23]. For the extraction, eutectic solvent composed of menthol and dodecanoic acid in a 3:1 M ratio was used, which had previously been identified as the most effective system for PDO extraction [24]. The initial molecular structures of menthol and dodecanoic acid (for the ES), along with the fermentation broth components including 1,3-propanediol (PDO), 2,3-butanediol (BDO), glycerol (GLY), succinic acid (SUC), lactic acid (LAC), and citric acid (CIT), were constructed using Avogadro (version 1.2.0), as illustrated in Fig. 1 [25]. Molecular dynamics (MD) simulations were performed using GROMACS 2024 [26].

Table 1
Composition of components in different cases for assessing MD (3:1) extraction performance.

	Components	Mass fraction	References
(Based on experimental composition)			
Case-1	1,3-PDO	0.14–0.16	Janković et al., 2024 (21)
	Glycerol	0.015–0.016	
	Water	0.79–0.84	
Case-2	1,3-PDO	0.11–0.15	Janković et al., 2024 (21)
	2,3-BDO	0.025–0.033	
	Succinic acid	0.007–0.0133	
	Water	0.78–0.83	
Case-3	1,3-PDO	0.22–0.24	Laura et al., 2020 (22)
	Lactic acid	0.015–0.0175	
	Citric acid	0.025–0.03	
	Succinic acid	0.028–0.032	
	Water	0.71–0.73	
Case-4	1,3-PDO	0.14–0.16	Dhakal & Shah, 2019 (23)
	2,3-BDO	0.02–0.03	
	Water	0.81–0.84	
(Designed systems: equal composition of PDO and co-solutes)			
Case-5	1,3-PDO	0.10	
	Glycerol	0.10	
	Water	0.80	
Case-6	1,3-PDO	0.10	
	2,3-BDO	0.10	
	Water	0.80	
Case-7	1,3-PDO	0.10	
	Succinic acid	0.10	
	Water	0.80	

The OPLS-AA (Optimized Potentials for Liquid Simulations - All Atom) force field was employed to describe bonded and non-bonded interactions, including bond stretching, angle bending, dihedral, van der Waals, and electrostatic interactions. The electrostatics were computed using Coulomb's law, while van der Waals interactions were modeled using a 12–6 Lennard-Jones potential. Cross-interactions were evaluated using standard Lorentz-Berthelot combination rules. Force field parameters for all feedstock components were generated using the LigParGen web server based on the 1.14*CM1A charge model [27–29] (Supplementary information A). Water molecules were modeled using the OPC3 (Optimal Point Charge) water model [30].

A cubic simulation box with dimensions of 70 × 70 × 70 Å^3 was constructed, containing 150 ES molecules (450 menthol and 150 dodecanoic acid). Energy minimization was initially performed, followed by a constant-volume (NVT) heating simulation from 5 K to 400 K over 500 ps. Temperature annealing from 400 K to 300 K was conducted over 10 ns to promote adequate mixing. Subsequently, the system was equilibrated at 298 K and 1 atm in the constant-pressure (NPT) ensemble for 20 ns for validation of the density of the ES. Following the equilibration of the pure ES phase, a two-phase liquid-liquid system was built by extending the simulation box in the x-direction and introducing aqueous feedstock solutions on either side of the ES phase.

Periodic boundary conditions were applied in all directions. Electrostatic interactions were calculated using the Particle Mesh Ewald (PME) method with a cutoff of 1.2 nm, and bond constraints were applied using the LINCS algorithm [31]. Equilibration of the composite systems was carried out in the NVT ensemble at 350 K for 500 ps, followed by NPT equilibration at 350 K and 1 bar for another 500 ps. The system was then cooled to 298 K, and a 50 ns production run was conducted at 298 K and 1 bar using the Nose-Hoover thermostat (coupling

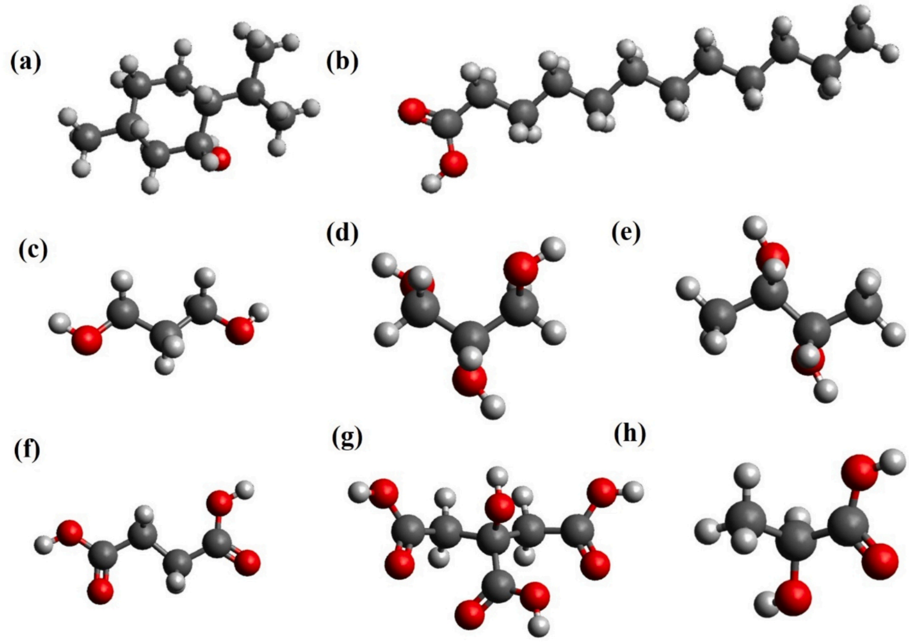

Fig. 1. Ball-and-stick molecular structures of (a) Menthol, (b) Dodecanoic acid, (c) 1,3-Propanediol (PDO), (d) Glycerol (Gly), (e) 2,3-Butanediol (BDO), (f) Succinic acid (SUC), (g) Citric acid (CIT), and (h) Lactic acid (LAC). Black spheres represent carbon atoms, grey spheres represent hydrogen atoms, and red spheres represent oxygen atoms. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

time 0.2 ps) [32,33] and Parrinello-Rahman barostat (coupling time 2.0 ps) [34] (*Details provided in supplementary information B*). Post-simulation analysis involved calculating density profiles along the x-axis to evaluate phase behavior and component distribution, while spatial distribution functions (SDFs) were generated using the TRAVIS package to visualize the three-dimensional arrangement of solutes within the system [35]. Visualization of structural features, solute orientation, and trajectory analysis was performed using VMD as part of the overall system characterization [36]. The extraction efficiency of each solute was determined by comparing its concentration in the ES and aqueous phases. Distribution coefficients (β) and the selectivity (S) of PDO over other components such as GLY, BDO, and organic acids were also computed. In addition, the interaction energies between ES and individual solutes were calculated to understand the molecular driving forces behind the separation. Finally, the self-diffusion coefficients of the solutes and ES were determined using the mean square displacement (MSD) method, providing insight into their dynamic mobility within the system.

2.2. Gibbs free energy of solvation

An alchemical free energy approach was employed to determine the solvation-free energy (ΔG_{solv}) of PDO, BDO, Gly, and SUC in water and MD (3:1) [37]. The calculations followed a two-stage coupling scheme, in which the Lennard-Jones interactions were introduced first, followed by the electrostatic interactions. This transformation was guided by a coupling parameter (λ) across 21 discrete steps, progressively transitioning the solute from a non-interacting to a fully interacting state. A soft-core potential with sc-power = 1, sc-sigma = 0.3, and sc-alpha = 1.0 was applied to prevent singularities.

Each solvation simulation involved a single solute molecule in 200 MD (3:1) or 3000 water molecules. The system was first allowed to undergo energy minimization, followed by equilibration in the NVT ensemble for 0.5 ns. Thereafter, equilibration of the system was performed in the NPT ensemble for 5 ns. For each λ value, 2 ns NPT equilibration was performed, followed by a 3 ns production run. The production runs for the free energy of solvation were carried out in the NPT ensemble, with temperature and pressure controlled using the Nosé-Hoover thermostat and Parrinello-Rahman barostat, respectively. The equation of motion was integrated using a stochastic dynamics integrator with a time step of 1 fs.

The free energy difference (ΔG_{solv}) between lambda states was calculated using the thermodynamic integration, given by eq. 1

$$\Delta G_{solv} = \int_0^1 \left\{ \frac{dH}{d\lambda} \right\} d\lambda \tag{1}$$

where H is the Hamiltonian, representing the energy function of the system. In GROMACS, the software outputs $\frac{dH}{d\lambda}$ (the derivative of the Hamiltonian with respect to λ) at each predefined λ state. To determine ΔG_{solv}, we integrate these values over the entire range of λ from 0 to 1, effectively summing the contributions from each step along the alchemical transformation path. Finally, the solvation free energy was determined using the Bennett acceptance ratio (BAR) method, as implemented in GROMACS [38] *(Additional details can be found in the Supplementary information C)*. These calculations provide insights into the solvation free energy of components in water and MD (3:1), helping to quantify the interaction strength between components and the surrounding solvent environment.

3. Results and discussions

3.1. Experimental density validation

To validate the reliability of the force field parameters, density calculations were done for all components and the ES (MD (3:1)) at 298 K and 0.1 MPa. The densities were evaluated using two different partial charge scaling factors, ±0.8 and ± 1.0. As presented in Table S2, the simulated densities obtained with ±0.8 charge scaling showed significantly closer agreement with experimental values across all compounds. The average absolute relative deviation (%AARD) was 0.581 % for the ±0.8 scaling, compared to 1.727 % for the ±1.0 scaling. For example, the simulated density of PDO was 1.052 g/cm^3 at ±0.8 scaling versus the experimental value of 1.049 g/cm^3, with only a 0.286 % deviation. Similarly, the ES showed a %ARD of 0.440 with ±0.8 scaling, as opposed to 1.671 with ±1.0 scaling. Based on these observations, all subsequent simulations used the ±0.8 charge scaling to represent system properties accurately.

3.2. Local mass density profile and extraction assessment

To evaluate the selective extraction behavior of the MD(3:1), we employed MD simulations to examine the spatial distribution and partitioning of PDO and various co-solutes in biphasic systems. The blue region in the simulation box represents the ES phase, predominantly occupying the central region, while the red regions at both ends correspond to the aqueous (water) phase, clearly demonstrating effective phase separation across all cases, as depicted in Fig. 2 for the experimental cases and Fig. S1 for the designed systems. In Fig. 2(a) and Fig. 2 (b), PDO (orange) is predominantly localized within the ES phase, confirming its strong affinity for the hydrophobic solvent. In contrast, Gly (green) in Fig. 2(a) and BDO (black) in Fig. 2(b) are mainly distributed near the interface and within the aqueous region, indicating their limited affinity toward the ES compared to PDO. Additionally, in Fig. 2(b), succinic acid (SUC, yellow) is almost entirely present in the aqueous phase, further highlighting the selectivity of the MD(3:1) for PDO over co-existing solutes. Fig. 2(c) shows a similar trend where PDO is transferred in the ES phase, while other organic acids such as SUC (yellow), LAC (grey), and CIT (purple) are predominantly localized in the aqueous phase. This distribution highlights the solvent's selective exclusion of these highly polar acids and further supports the preferential partitioning of PDO into the hydrophobic ES. In Fig. 2(d), the strong partitioning of PDO into the ES is once again observed, while BDO remains mostly at the interface, with minimal penetration into the ES. Fig. S1 demonstrates that even under designed conditions with equal mass concentrations of PDO and impurities, PDO still preferentially partitions into the ES phase, indicating a stronger affinity than the other solutes.

Together, these local mass density profiles highlight the exceptional selectivity of the MD (3:1) for PDO, while effectively rejecting water and other polar solutes commonly found in fermentation broth. These qualitative observations are further corroborated by the quantitative analyses of distribution coefficients (β), selectivity (S), and extraction efficiency (E).

The distribution component (β) was calculated using Eq. 2, where m_i^E and m_i^R are the mass fractions of component i in the extract (ES) and raffinate (aqueous) phases, respectively. Selectivity determinations were done using Eq. 3, where β_i and β_j denote the distribution coefficient of the target compound (PDO) and co-existing impurities, respectively. In addition, the extraction efficiency was determined by Eq. 4, where β_{ij} is the distribution coefficient of solute i in the solvent j.

$$\text{Distribution coefficient } (\beta_i) = \frac{m_i^E}{m_i^R} \tag{2}$$

$$\text{Selectivity } (S_{ij}) = \frac{\beta_i}{\beta_j} \tag{3}$$

$$\%\text{Extraction efficiency } (E) = \left[\frac{\beta_{ij}}{1+\beta_{ij}}\right] \times 100 \tag{4}$$

Tables S3 and S4 present the molecules distribution and mass composition, respectively, of feedstock components between the extract and raffinate phases after a 50 ns molecular dynamics simulation at 298 K. Table 2 summarizes the calculated distribution coefficients of different components and selectivity for PDO across seven cases, including four experimental setups (Cases 1–4), which mimic real fermentation broths, and three designed systems (Cases 5–7).

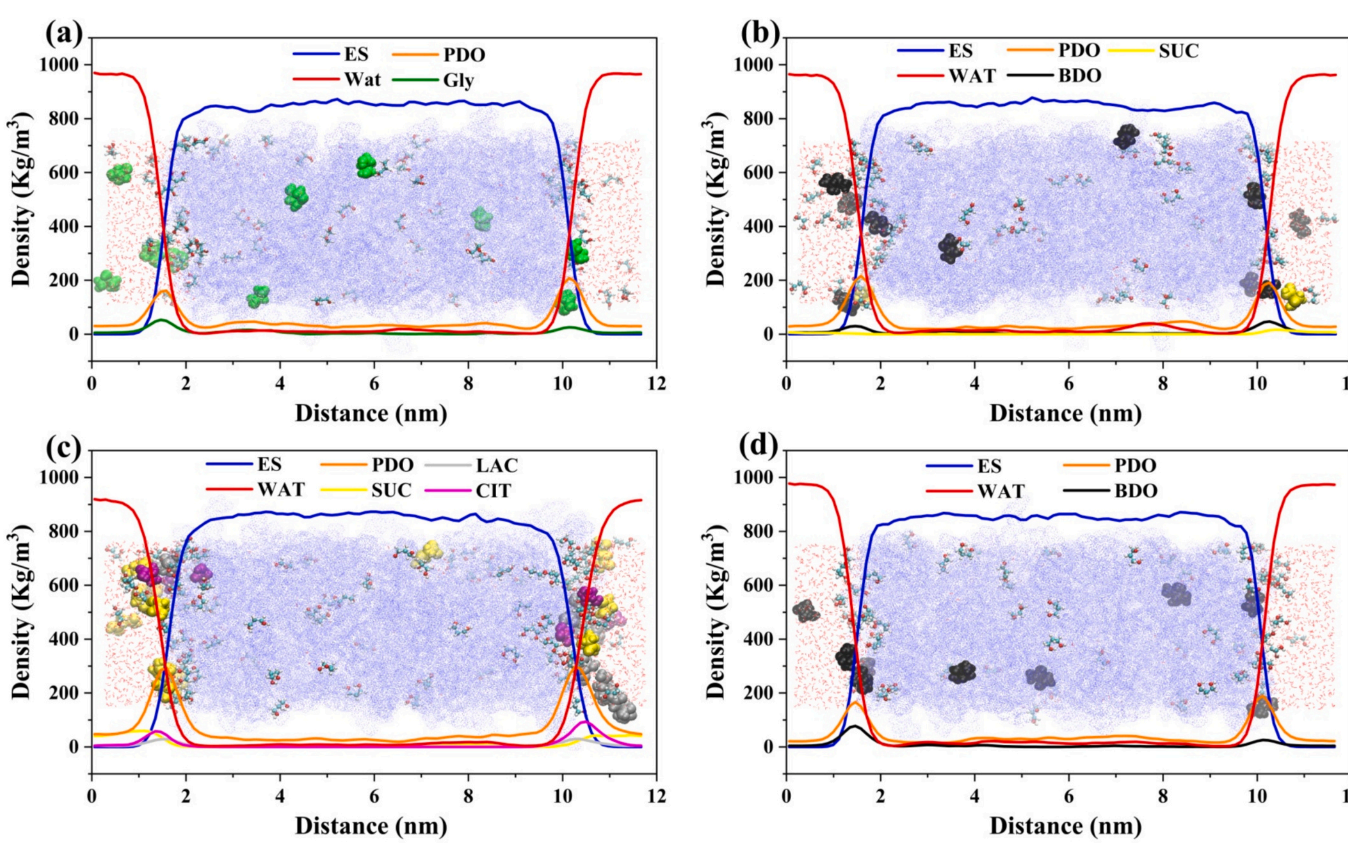

Fig. 2. Local mass density profile of different components of fermentation broth at $T = 298$ K and $p = 0.1$ MPa. Blue colour denotes ES phase, red colour denotes aqueous phase, green colour is glycerol, black is BDO, yellow is succinic acid, grey is lactic acid, orange is PDO and purple is citric acid. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Mass distribution coefficient and selectivity of components in different cases at $T = 298$ K and $p = 0.1$ MPa.

	Distribution coefficient (β)					Selectivity (S)			
CASE 1	β_{PDO}	β_{Gly}	β_{Wat}			$S_{PDO/GLY}$	$S_{PDO/Wat}$		
	4.333	0.600	0.032			7.22	135.41		
CASE 2	β_{PDO}	β_{BDO}	β_{SUC}	β_{Wat}		$S_{PDO/SUC}$	$S_{PDO/BDO}$	$S_{PDO/Wat}$	
	2.932	0.391	0.000	0.043		∞	7.50	68.19	
CASE 3	β_{PDO}	β_{SUC}	β_{LAC}	β_{CIT}	β_{Wat}	$S_{PDO/SUC}$	$S_{PDO/LAC}$	$S_{PDO/CIT}$	$S_{PDO/Wat}$
	2.984	0.083	0.176	0.000	0.030	35.95	16.95	∞	99.47
CASE 4	β_{PDO}	β_{BDO}	β_{Wat}			$S_{PDO/BDO}$	$S_{PDO/Wat}$		
	4.429	0.600	0.044			7.38	100.66		
CASE 5	β_{PDO}	β_{Gly}	β_{Wat}			$S_{PDO/GLY}$	$S_{PDO/Wat}$		
	2.656	0.690	0.040			3.85	66.40		
CASE 6	β_{PDO}	β_{BDO}	β_{Wat}			$S_{PDO/BDO}$	$S_{PDO/Wat}$		
	4.091	0.671	0.041			6.10	99.78		
CASE 7	β_{PDO}	β_{SUC}	β_{Wat}			$S_{PDO/SUC}$	$S_{PDO/Wat}$		
	2.833	0.160	0.029			17.71	97.69		

As shown in Table S3, MD(3:1) remained in the solvent phase across all the cases, whereas several PDO molecules transferred from the aqueous to the solvent phase. Additionally, a few water molecules had moved to the extract phase, with their composition ranging from 2.5 wt % to 4.2 wt%. In our previous work, where we reported the system consisting of only PDO in the aqueous solution, the water composition in the extract phase was only 1.3 wt% (20). This indicates that the presence of other components in the aqueous solution facilitates more water transfer to the extract phase, which in turn will slightly reduce the selectivity of MD (3:1) for PDO over water. However, this observed increase is nominal and will not significantly impact the solvent's effectiveness. Other components, such as GLY and BDO, have also shown some interaction with the solvent. However, the number of molecules transferred is less, with a composition of less than 1.2 wt% in the extract phase (cases 1–4). Acids such as succinic, citric, and lactic acid have shown almost negligible interaction with the solvent and remained mainly in the aqueous solution, with composition in the extract phase less than 0.5 wt% (cases 1–4). On comparing case-1 (PDO:16 wt%, Gly: 3 wt%) to case-5 (PDO:10 wt%, Gly: 10 wt%), we observe that PDO composition in the extract phase reduced from 14.3 wt% to 8.5 wt%, whereas GLY composition increased from 1.2 wt% to 4.9 wt%. Similarly, PDO composition in the extract phase reduced from 12.4 wt% to 9.0 wt %, whereas BDO composition increased from 1.2 wt% to 4.7 wt% (case-4 vs case-6). The above analysis showed the impact of the components present in the aqueous phase on the extraction performance of the solvent toward PDO. However, the solvent still prefers PDO over GLY and BDO.

As shown in Table 2, PDO exhibited significantly higher distribution coefficients (β), ranging from 2.932 to 4.429 across cases 1–4, compared to the co-solutes. GLY and BDO both showed lower β values of approximately 0.60. In contrast, organic acids such as SUC and LAC had β values below 0.2, showing minimal extraction into the solvent phase. On comparing case-1 (PDO: 16 wt%, Gly: 3 wt%) to case-5 (PDO: 10 wt%, Gly: 10 wt%), we observe that β_{PDO} reduced from 4.333 to 2.656, whereas β_{Gly} slightly increased from 0.60 to 0.69. It suggests that GLY composition affects the extraction of PDO, but still, β_{PDO} is much larger than β_{Gly}. Similarly, β_{PDO} reduced from 4.429 to 4.091, whereas β_{BDO} increased from 0.60 to 0.671 (case-4 vs case-6). But this change is not very significant. The analysis of β values indicates that $\beta_{PDO,}$ even in the presence of other components, is much larger, suggesting more solvent affinity toward PDO than other co-solutes.

High selectivity values for PDO compared to water were observed for cases 1–4, ranging from 68.19 to 135.41. Similarly, high selectivity for PDO over other co-solutes was observed in cases 1–4, reflecting the remarkable preference of the MD(3:1) for extracting PDO over water and other impurities. Conversely, the designed systems showed slightly reduced selectivity with $S_{PDO/wat}$ ranging from 66.40 to 99.78. Selectivity values for PDO over GLY, BDO, and SUC were 3.85, 6.10, and 17.71. Despite the uniform starting concentrations, MD (3:1) maintained a favorable extraction performance for PDO, highlighting its intrinsic affinity toward the target compound.

To further assess solvent performance, the extraction efficiency of each component was evaluated, as summarized in Table 3. In cases 1–4, PDO extraction efficiencies were consistently high, ranging from 75 % to 82 %, affirming the solvent's ability to selectively and efficiently extract the target compound under realistic conditions. In contrast, water and other solutes such as Gly, BDO, SUC, LAC, and CIT exhibited significantly lower extraction efficiencies, generally below 40 % and less than 4 % for water, indicating minimal co-extraction and further validating the hydrophobic and selective nature of MD (3:1). In the designed systems, where equal mass fractions were used, PDO extraction efficiencies remained within a comparable range (73–80 %), suggesting that the solvent's selective behavior is not solely influenced by feed composition but is also inherent to the molecular interactions between PDO and the MD (3:1). Overall, these findings demonstrate the high selectivity and extraction efficiency of MD (3:1) for PDO, even under complex, fermentation-like conditions. Moreover, the comparative analysis between experimental and designed systems underscores the importance of feed composition in evaluating solvent performance for practical applications.

3.3. Hydrogen bonding

Hydrogen bonding plays a critical role in governing solute-solvent interactions, significantly influencing the selectivity and efficiency of liquid-liquid extraction processes. To accurately evaluate these interactions, molecules were considered to form a hydrogen bond when the donor-acceptor distance was below 3.5 Å and the hydrogen bond

Table 3
Extraction efficiency (%) of different components into the solvent phase (ES) at $T = 298$ K and $p = 0.1$ MPa.

	Extraction Efficiency (%)				
CASE 1	E_{PDO}	E_{Gly}	E_{Wat}		
	81.00	38.00	3.00		
CASE 2	E_{PDO}	E_{BDO}	E_{SUC}	E_{Wat}	
	75.00	28.00	0.00	4.00	
CASE 3	E_{PDO}	E_{SUC}	E_{LAC}	E_{CIT}	E_{Wat}
	75.00	8.00	15.00	0.00	3.00
CASE 4	E_{PDO}	E_{BDO}	E_{Wat}		
	82.00	38.00	4.00		
CASE 5	E_{PDO}	E_{Gly}	E_{Wat}		
	73.00	41.00	4.00		
CASE 6	E_{PDO}	E_{BDO}	E_{Wat}		
	80.00	40.00	4.00		
CASE 7	E_{PDO}	E_{SUC}	E_{Wat}		
	74.00	14.00	3.00		

angle deviated no more than 30° from linearity [39,40]. These geometric criteria ensured consistent identification of hydrogen bonding across all simulated systems. The bar chart in Fig. 3 illustrates the average number of hydrogen bonds formed between solutes and MD 3:1 in the extract phase and between solutes and water in the raffinate phase, across all cases. This data was extracted from Fig. S2, which shows the time evolution of hydrogen bonding interactions over a 5000 ps simulation period. For quantification, the final 1000 ps of each trajectory were segmented into three equal parts, and the average number of hydrogen bonds was calculated from these segments to ensure representative sampling of equilibrated interactions. In each case, the first series of bars (left of the gap) represents hydrogen bonding between different components and MD (3:1), and the second series of bars (right of the gap) corresponds to hydrogen bonding between different components and water. The data for this chart was derived from Fig. S2 of the supplementary section.

Across all systems, PDO forms the highest number of hydrogen bonds with MD (3:1). This strong interaction results from PDO's two hydroxyl (-OH) groups, which act as hydrogen bond donors and acceptors. These groups allow PDO to form directional and cooperative hydrogen bonds with both components of the MD (3:1). It works as a hydrogen bond donor for the carbonyl oxygen of dodecanoic acid (a hydrogen bond acceptor) and a hydrogen bond acceptor from the hydroxyl group of menthol (a hydrogen bond donor). This dual capability produces a highly stable hydrogen-bonding network, reinforcing PDO's preferential partitioning into the extract phase. In contrast, organic acids such as CIT, SUC, and LAC also possess polar (-OH) and (-COOH) groups. However, they form negligible hydrogen bonds with MD (3:1). Instead, they exhibit strong hydrogen bonding with water, particularly in cases 2 and 3 and designed case-7. Their strong hydration, especially for polyfunctional acids like CIT with multiple -COOH groups, prevents effective interaction with the hydrophobic ES, leading to their retention in the aqueous raffinate phase and explaining their low distribution coefficients. Gly and BDO, which also contain multiple -OH groups and are hydrophilic like PDO, exhibit some hydrogen bonding with the ES across all cases. However, such interactions are significantly lower than those observed for PDO. While Gly and BDO are capable of interacting with the ES, their -OH groups show a greater tendency to form hydrogen bonds with water. As a result, they are mainly retained in the aqueous phase rather than PDO. This highlights that although Gly and BDO share similar polarity characteristics with PDO, the orientation and strength of their hydrogen bonding with ES components are less favorable, limiting their extraction efficiency.

The aqueous phase behavior further supports these observations. In relevant cases, organic acids form over five hydrogen bonds (per organic acid molecule) with water, emphasizing their high aqueous solubility. PDO forms fewer hydrogen bonds in the raffinate phase than other solutes in all cases, indicating less retention in water and a stronger pull toward the extract phase. Gly and BDO also show higher hydrogen bonding in water than PDO across all the cases, validating their low extraction behavior. When comparing real (Cases 1–4) and designed (Cases 5–7) systems, the observed trends remain consistent in all cases. PDO exhibits dominant hydrogen bonding with the ES and limited interaction with water, supporting its highly selective extraction. Organic acids are mainly retained in the aqueous phase due to extensive hydration. At the same time, Gly and BDO are extracted to a lesser extent than PDO due to their comparatively weaker interactions with MD (3:1). These findings reinforce that extractability is not solely determined by hydrophilicity but by the specific ability of solutes to engage in cooperative hydrogen bonding with both the hydrogen bond donor and acceptor components of the eutectic solvents. The MD (3:1) thus emerges as a selective and effective medium for isolating PDO from multi-component aqueous systems.

To further investigate the system's ability to capture strong hydrogen bonding interactions, combined distribution functions (CDFs) were generated between the ES and PDO to validate the applied geometric criteria [35,41]. These CDFs integrate radial distribution functions (RDFs) and angular distribution functions (ADFs), offering detailed spatial and orientational insights. The radial component represents the distance between the hydroxyl (-OH) oxygen of the reference molecule and the oxygen atom of the observed molecule, while the angular component corresponds to the angle formed by the hydrogen donor and acceptor atoms. These analyses were specifically carried out for the designed systems in Case 5, Case 6, and Case 7.

Two interaction geometries were considered in the analysis. In the first case, the distance between the -OH oxygen of menthol and the oxygen atom of PDO was measured, along with the corresponding angle formed by the hydrogen of menthol, the oxygen of menthol, and the oxygen of PDO. In the second case, the distance between the -OH oxygen of PDO and the oxygen atom of menthol was analyzed, along with the corresponding angle formed by the hydrogen of PDO, the oxygen of PDO, and the oxygen of menthol. As presented in Fig. S3 of the Supplementary Information, hydrogen bonding in all systems was found to be most prominent at distances less than 3.5 Å and angles below 30°, satisfying the geometric criteria for hydrogen bond formation.

3.4. Interaction energy

The interaction energies between the solvent MD (3:1) and the solutes were calculated using molecular dynamics simulations to understand further the molecular driving forces governing selective extraction. Interaction energy is a quantitative indicator of the affinity

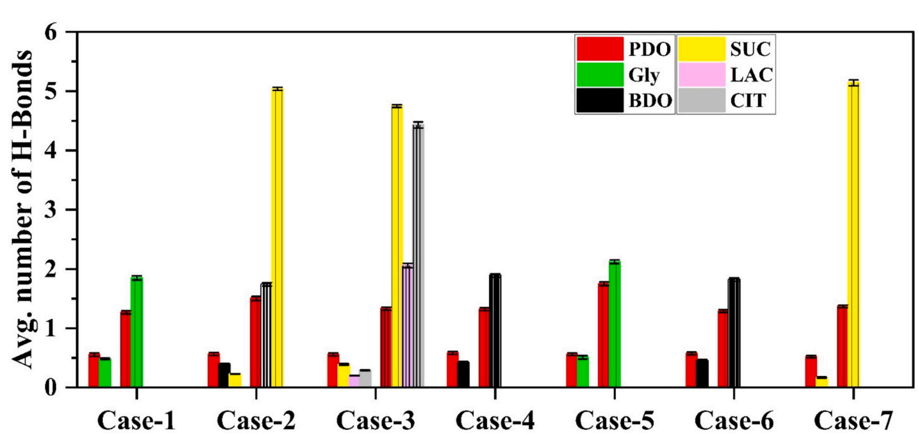

Fig. 3. Average number of hydrogen bonds (per solute/co-solute) between ES or water and different components of fermentation broth. The first set of bars in each case represents hydrogen bonding between the ES and compounds in the extract phase, while the second set of bars represents hydrogen bonding between water and components in the raffinate phase.

between molecules, reflecting the strength and nature of intermolecular interactions that contribute to solute partitioning. This study computed the total interaction energy as the sum of van der Waals and Coulombic interactions, as detailed in Table S5 (Supplementary Information) and summarized in Table 4. This analysis explains why PDO preferentially partitions into the extract phase compared to other co-existing solutes. More negative values indicate stronger attractive interactions, which correlate with higher solubility or preferential localization in a particular phase.

In all cases, PDO exhibited significantly more favorable interaction energies with MD (3:1) compared to most co-solutes, confirming its strong preferential affinity for the solvent. For example, in Case 1, the ES-PDO interaction energy is −20.88 kJ/mol, which is substantially more negative than that of GLY (−3.45 kJ/mol), highlighting the ES's selective interaction with PDO. Similar trends are observed in Case 2, where the ES-PDO interaction (−18.75 kJ/mol) is stronger than with BDO (−3.03 kJ/mol) and SUC (−0.16 kJ/mol), underscoring the affinity-driven preference for PDO in the extract phase. In Case 3, organic acids such as SUC, LAC, and CIT exhibit weak interaction energies with the ES (−1.32, −0.98, and − 1.61 kJ/mol, respectively), while the ES-PDO interaction remains relatively strong (−24.82 kJ/mol), indicating that the solvent disfavors the extraction of polar acids while maintaining a strong interaction with PDO. Case 4 further supports these findings, where the ES-PDO interaction energy (−17.48 kJ/mol) is more favorable than that of BDO (−2.79 kJ/mol), reaffirming the solvent's selective affinity.

The designed systems (Cases 5–7), which account for an equal composition of all solutes to simulate competitive extraction scenarios, further confirm this selective behavior. In Case 5, the ES-PDO interaction energy (−11.28 kJ/mol) is stronger than with GLY (−9.82 kJ/mol), while in Case 6, PDO (−12.79 kJ/mol) again exhibits more favorable interaction than BDO (−11.42 kJ/mol). Similarly, in Case 7, the ES-PDO interaction (−12.69 kJ/mol) is significantly more negative than that with SUC (−1.44 kJ/mol). Despite the general tendency of water to form hydrogen bonds and interact with polar solutes, the ES maintains its preferential interaction with PDO across all cases, indicating robust selectivity even under uniformly distributed solute conditions. This trend indicates that PDO is still preferentially extracted under conditions with uniform feed compositions into the extract phase, further validating the selectivity toward MD (3:1).

Table 4
Interaction energies (in kJ/mol) between ES or water and various solutes in experimental (Cases 1–4) and designed (Cases 5–7) systems at $T = 298$ K and $p = 0.1$ MPa. Interaction energy is based on per ES or water molecule.

		Interaction Energy (KJ/mol)		Interaction Energy (KJ/mol)
CASE 1	ES - PDO	−20.882	Wat - PDO	−2.159
	ES - Gly	−3.447	Wat - Gly	−0.442
CASE 2	ES - PDO	−18.747	Wat - PDO	−2.660
	ES - BDO	−3.026	Wat - BDO	−0.521
	ES - SUC	−0.156	Wat - SUC	−0.261
CASE 3	ES - PDO	−24.823	Wat - PDO	−3.703
	ES - SUC	−1.320	Wat - SUC	−1.664
	ES - LAC	−0.981	Wat - LAC	−0.431
	ES - CIT	−1.611	Wat - CIT	−0.911
CASE 4	ES - PDO	−17.483	Wat - PDO	−1.952
	ES - BDO	−2.791	Wat - BDO	−0.490
CASE 5	ES - PDO	−11.282	Wat - PDO	−1.702
	ES - Gly	−9.820	Wat - Gly	−2.028
CASE 6	ES - PDO	−12.797	Wat - PDO	−1.380
	ES - BDO	−11.415	Wat - BDO	−1.731
CASE 7	ES - PDO	−12.689	Wat - PDO	−1.406
	ES - SUC	−1.441	Wat - SUC	−2.211

These interaction energy trends are consistent with the findings of the extraction efficiency and hydrogen bond analysis. The consistently stronger interactions of PDO with the ES compared to other fermentation broth components highlight the importance of electrostatic compatibility and hydrophobic interactions in driving the selective extraction. The significantly weaker interactions of the ES with organic acids and polar alcohols, such as GLY and lactic acid, explain their exclusion from the ES phase and support the observed high selectivity of the MD (3:1) for PDO. This analysis provides valuable mechanistic insight into the molecular selectivity of ESs. It could guide the design and optimization of ESs for bioproduct separations, ensuring more efficient and targeted extraction processes.

3.5. Structural analysis

To further support the understanding of the selective extraction of PDO, spatial distribution function (SDF) calculations were performed. In Cases 1 to 4, the SDF plots (Fig. 4) display concentrated red isosurfaces around the hydroxyl (-OH) group of menthol and the carboxyl (-COOH) group of dodecanoic acid, indicating a strong spatial preference of PDO for these functional groups. This structural distribution aligns well with the hydrogen bonding analysis, confirming that PDO is present in significant quantity around the MD (3:1) components due to specific intermolecular interactions with the hydroxyl and carboxyl functional groups. In contrast, co-solutes such as GLY, BDO, and organic acids (SUC, CIT, LAC) in Cases 2 and 3 exhibit significantly lower or negligible spatial densities around the ES components, reflecting their limited interaction and poor extractability. Supplementary Fig. S4 further supports this observation by showing that organic acids are either sparsely present or absent around ES in cases 2 and 3.

To further evaluate the performance of the solvent under competitive extraction conditions, multi-component systems were examined through cases 5 to 7. The SDF plots for these cases, shown in Supplementary information Fig. S5 (isovalue = 0.55), again revealed that PDO maintains a strong spatial association with the ES components. In contrast, GLY, BDO, and SUC exhibited minimal density in the ES region. In the corresponding aqueous (raffinate) phase, depicted in the rightmost panels of each row, water demonstrates stronger spatial associations with GLY, BDO, and SUC compared to PDO. It suggests that water preferentially retains these co-solutes, thereby impeding their partitioning into the ES phase and enhancing the selective extraction of PDO. Taken together, these spatial distribution insights, along with the hydrogen bonding analyses, provide robust evidence of the high specificity and efficiency of the MD (3:1) solvent in selectively extracting PDO over other co-occurring solutes.

3.6. Self-diffusion coefficient

To understand the molecular-level transport behavior and validate the selective extraction capability of the solvent, we investigated the diffusion dynamics of all solutes within the extract phase by analyzing their mean square displacement (MSD) and calculating the corresponding self-diffusion coefficients. The MSD and diffusion coefficient were determined using Eqs. 5 and 6, where $r(t)$ and $r(0)$ represent the center of mass position of each particle at time t and time 0, respectively;

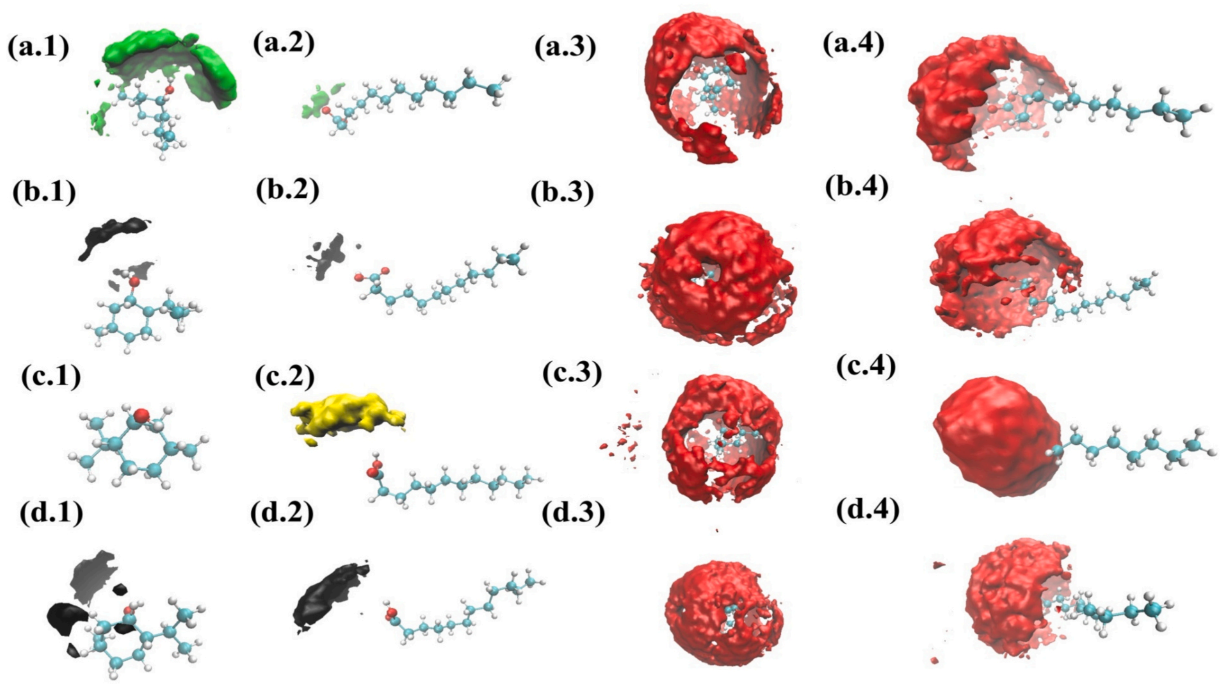

Fig. 4. Spatial distribution function (SDF) isosurfaces for the extraction systems studied in Cases 1 to 4, illustrating the localization of solute molecules around the components of the ES. Each row (a–d) corresponds to a specific extraction case: (a) Case 1, (b) Case 2, (c) Case 3, and (d) Case 4. Columns 1 and 2 show the SDF isosurfaces of co-solutes around menthol and dodecanoic acid, respectively, while columns 3 and 4 (x.3 and x.4) display the corresponding isosurfaces for PDO in the same cases. Colour coding denotes different solutes: green for GLY, black for BDO, yellow for SUC, and red for PDO. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

N is the total number of particles; the angular brackets [] denote ensemble averaging; and the factor 1/6 accounts for diffusion in three-dimensional space.

$$MSD = \frac{1}{N}\left[\sum_{i=1}^{N} |r_i(t) - r_i(0)|^2\right] \tag{5}$$

$$D_s = \frac{1}{6}\lim_{n\to\infty}\frac{d}{dt}\left(\sum_{i=1}^{N} |r_i(t) - r_i(0)|^2\right) \tag{6}$$

Fig. S6 presents the MSD plots of various solutes in the extract phase, reflecting their diffusion behavior in cases 1–4. Generally, a steeper MSD slope indicates higher molecular mobility and thus weaker interactions with the solvent, while a less steep slope corresponds to slower diffusion, suggesting stronger interactions or solubility affinity with the solvent. Water consistently exhibited the steepest MSD slope and highest mobility among all solutes, highlighting its weak interaction with the hydrophobic ES and confirming its preference for the aqueous phase. In contrast, PDO showed relatively lower mobility across all cases, as indicated by its less steep MSD slope, implying stronger interaction and higher solubility affinity with the ES. This observation aligns with the experimentally observed selective extraction of PDO. Co-solutes such as Gly and BDO show higher MSD slopes than PDO, reflecting greater mobility and thus weaker interactions with the ES, which explains their lower extraction efficiency. Organic acids such as SUC, CIT, and LAC are present in only trace amounts in realistic extraction scenarios. Their MSD plots exhibit relatively low slopes, suggesting limited diffusion. However, due to their minimal concentrations, these low diffusion rates could result from statistical uncertainty rather than true interaction strength. To overcome this limitation and validate the observed trends, we designed controlled systems in cases 5–7, where PDO and co-solutes were introduced in equal mass ratios. This allowed for directly comparing competitive diffusion behavior under consistent compositional conditions. In these controlled systems, the MSD plots (Fig. S7 (Supplementary information)) clearly showed that PDO consistently exhibited a less steep slope compared to GLY, BDO, and SUC, reinforcing its relatively stronger binding and affinity toward the ES. This trend further substantiates the selective nature of the ES for PDO even in the presence of competing solutes.

To evaluate molecular mobility, the MSD curves for each component were first fitted to a linear region, and the quality of these fits was assessed using R^2 values, as listed in Table S6 (Supplementary information). Most solutes exhibited excellent linearity ($R^2 > 0.98$), indicating reliable statistical convergence even for components at relatively low concentrations. Based on the slope of these fitted MSD curves, the self-diffusion coefficients (D) were then calculated using Eq. 6. The resulting values are reported in Table 5. Across all systems, water exhibited the highest diffusion coefficients (ranging from 0.951 to 1.235 $\times 10^{-5}$ cm^2/s), reflecting its weak interaction with the ES and strong tendency to remain in the aqueous phase. In contrast, PDO showed consistently lower diffusion coefficients (0.169 to 0.304), reaffirming its stronger interaction and higher solubility affinity with ES. Gly and BDO generally exhibited slightly higher diffusivity than PDO, indicating weaker ES interactions. The organic acids (SUC, CIT, LAC) displayed variable diffusion values; however, due to their low concentrations, the results, particularly for SUC in case 2, carry high uncertainty. To eliminate this concentration bias, designed systems (cases 5–7) were constructed with equal PDO and co-solutes mass ratios. Even under these controlled conditions, PDO showed lower diffusivity than co-solutes like GLY, BDO, and SUC, confirming the ES's selective affinity and stronger binding interactions with PDO. In conclusion, the diffusion analysis confirmed that the ES exhibited a stronger molecular affinity and solubility preference for PDO than other co-solutes. This selective interaction underpins the effectiveness of the ES in extracting PDO from complex fermentation broth.

Table 5
Self-diffusion coefficient of components in the extract phase for different cases at $T = 298$ K and $p = 0.1$ MPa.

Self-diffusion coefficient (D) (10^{-5} cm^2/s)					
Case - 1	PDO	Gly	Wat		
	0.304 ± 0.040	0.371 ± 0.129	1.226 ± 0.028		
Case - 2	PDO	BDO	SUC	Wat	
	0.188 ± 0.0553	0.262 ± 0.060	0.048 ± 0.158	1.139 ± 0.013	
Case - 3	PDO	CIT	LAC	SUC	Wat
	0.238 ± 0.022	0.109 ± 0.075	0.195 ± 0.068	0.316 ± 0.044	0.951 ± 0.036
Case - 4	PDO	BDO	Wat		
	0.243 ± 0.097	0.253 ± 0.045	1.208 ± 0.039		
Case - 5	PDO	Gly	Wat		
	0.246 ± 0.055	0.279 ± 0.039	1.235 ± 0.027		
Case - 6	PDO	BDO	Wat		
	0.169 ± 0.017	0.228 ± 0.017	1.128 ± 0.027		
Case - 7	PDO	SUC	Wat		
	0.173 ± 0.031	0.252 ± 0.081	0.916 ± 0.046		

3.7. Gibb's free energy of solvation

To further substantiate the selective extraction behavior of the hydrophobic ES toward PDO, we evaluated the Gibbs free energy of solvation (ΔG_{solv}) for all major components, such as PDO, BDO, GLY, and SUC in the solvent and water phases, as presented in Fig.5 in section 3.7. The ΔG_{solv} serves as a thermodynamic indicator of solute-solvent affinity, where a more negative value signifies a greater favourability for solvation. Among all the solutes, PDO exhibited the most negative ΔG_{solv} in the solvent (−38.76 ± 1.89 kJ/mol), which is substantially lower than its value in water (−18.41 ± 1.83 kJ/mol), highlighting a strong preferential interaction of PDO with the solvent.

The ΔG_{solv} values of BDO and GLY are relatively similar in water and solvent. SUC displays a much more favorable solvation in water (−55.81 ± 2.44 kJ/mol) than in ES (−31.46 ± 2.36 kJ/mol), demonstrating its exclusion from the ES phase.

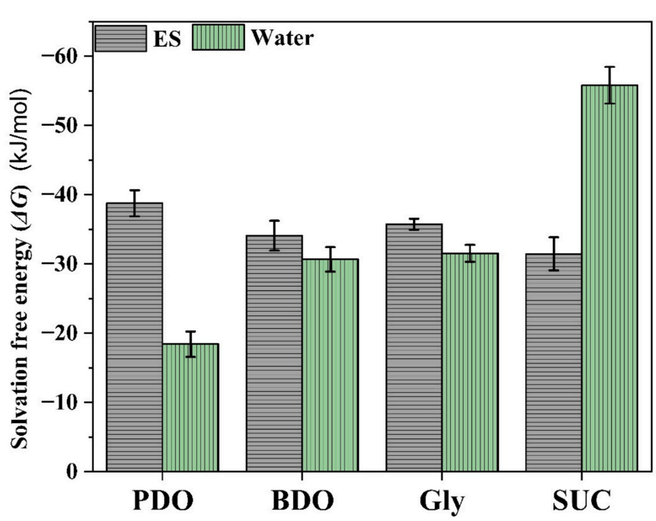

Fig. 5. Solvation free energy (ΔG) of different components of fermentation broth in ES and water at $T = 298$ K and $p = 0.1$ MPa.

The free energy of transfer of components from the aqueous phase to the eutectic solvent phase ($\Delta G_{transfer}$) can be evaluated using eq. 7, given by

$$\Delta G_{transfer} = \Delta G_{solv}^{ES} - \Delta G_{solv}^{water} \tag{7}$$

where, ΔG_{solv}^{ES} and ΔG_{solv}^{water} represent the free energy of solvation of components in eutectic solvent and water, respectively. The calculated values of $\Delta G_{transfer}$ represents the Gibbs free energy required for transferring the solute from the aqueous phase to the solvent phase. A negative value indicates that the transfer process is spontaneous. Among the studied compounds, $\Delta G_{transfer}$ follows the order: PDO (−20.35 kJ/mol) > GLY (−4.19 kJ/mol) > BDO (−3.42 kJ/mol), while SUC exhibits a positive value (+24.35 kJ/mol). The more negative value for PDO suggests stronger interactions with the eutectic solvent, indicating a higher affinity than GLY and BDO. In contrast, the positive value for SUC implies that it prefers to remain in the aqueous phase rather than partition into the solvent phase.

The partition coefficient of components can also be calculated with the free energy of transfer using eq. 8, given by

$$\text{Partition Coefficient}, \log(P) = \frac{-\Delta G_{transfer}}{2.303 \times \mathrm{R} \times T} \tag{8}$$

where R and T represent the gas constant and temperature, respectively.

The partition coefficient provides a comparative idea of the solubility of any compounds in two different immiscible phases. Similar to the free energy of transfer, the partition coefficient follows the order: PDO (3.57) > GLY (0.73) > BDO (0.60), while SUC exhibits a negative value (−4.26). These thermodynamic trends are consistent with the observed extraction results, supporting the conclusion that the eutectic solvent selectively extracts PDO. The enhanced affinity of water toward GLY, BDO, and SUC, as supported by Supplementary Fig. S4, further explains their retention in the aqueous phase, thereby improving the selectivity of PDO extraction. Overall, the solvation free energy analysis highlights the strong affinity of the eutectic solvent MD (3:1) for PDO over other solutes under the investigated conditions.

4. Conclusions

This study focused on the selective extraction of PDO from fermentation broth using a eutectic solvent composed of Menthol: Dodecanoic acid (3:1). Molecular dynamics simulations were used to evaluate the effectiveness of the solvent in the presence of several other co-solutes mimicking the experimental conditions. Three cases were specifically designed to assess the competitive affinity of major co-solutes that could potentially hinder the selective extraction of PDO. These additional cases were crucial in understanding how other fermentation broth by-products, such as GLY and BDO, might preferentially affect the ES's ability to extract PDO. The density profile indicated the formation of two phases, with no solvent loss and minimal water entry into the ES phase. The co-solutes, such as GLY and BDO, showed minimal entry into the ES phase, while organic acids exhibited almost no solubility. This was further supported by the selectivity and extraction efficiency data, where the extraction efficiency of PDO remains consistently around 80 % in all studied cases. In contrast, the entry of major co-solutes like BDO and GLY is limited to a maximum of 41 %, demonstrating the ES's selective extraction behavior. Interaction energy analysis highlighted stronger interactions between PDO and ES than other fermentation broth components. Structural analysis, including hydrogen bond analysis and the radial distribution function (SDF), further supported this. The ES-PDO interaction is characterized by a higher average number of hydrogen bonds, particularly between PDO and the -OH group of Menthol and Dodecanoic acid, indicating a strong orientation and favorability of PDO extraction by the ES. Conversely, less hydrogen bonding is observed between water and PDO, confirming the selective nature of the extraction. Diffusivity analysis through MSD slope revealed that PDO exhibited a lower self-diffusion coefficient than other co-solutes, particularly in controlled cases, reflecting the strong interaction between PDO and ES. The Gibb's free energy of solvation (ΔG) analysis revealed that ES preferentially interacted with PDO over other co-solutes, as indicated by more negative ΔG values for PDO in ES. Overall, the present study highlighted the effectiveness of the eutectic solvent MD (3:1) to selectively extract PDO while limiting the entry of other impurities, as water holds onto the co-solutes, ensuring their minimal interference in the extraction process. To enable industrial-scale implementation, which is the need of the hour for the advancing biorefinery sector, further research is needed to study the stability and recyclability of the eutectic solvent.

CRediT authorship contribution statement

Raj Akshat: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kanchan Kumari:** Investigation, Formal analysis. **Anand Bharti:** Writing – review & editing, Supervision, Software, Resources, Methodology, Formal analysis, Conceptualization. **Padmini Padmanabhan:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2025.128460.

Data availability

Data will be made available on request.

References

[1] A. Kamboj, et al., Unravelling the potential of sugarcane bagasse: an eco-friendly and inexpensive agro-industrial waste for the production of valuable products using pretreatment processes for sustainable bio-economy, J. Environ. Chem. Eng. 12 (6) (2024) 114461, https://doi.org/10.1016/j.jece.2024.114461.

[2] A. Chandel, S. da Silva, W. Carvalho, O. Singh, Sugarcane bagasse and leaves: foreseeable biomass of biofuel and bio-products, J. Chem. Technol. Biotechnol. 87 (Apr. 2012), https://doi.org/10.1002/jctb.2742.

[3] Anu Alokika, A. Kumar, V. Kumar, B. Singh, Cellulosic and hemicellulosic fractions of sugarcane bagasse: potential, challenges and future perspective, Int. J. Biol. Macromol. 169 (2021) 564–582, https://doi.org/10.1016/j.ijbiomac.2020.12.175.

[4] 1,3-Propanediol Market Size, Share Growth 2024–-2030, Accessed: May 28. [Online]. Available: https://www.precisionbusinessinsights.com/market-reports/1-3-propanediol-market, 2025.

[5] J.V. Kurian, A new polymer platform for the future - Sorona® from corn derived 1,3-propanediol, J. Polym. Environ. 13 (2) (Apr. 2005) 159–167, https://doi.org/10.1007/S10924-005-2947-7/METRICS.

[6] S. Anand, K. Mukherjee, P. Padmanabhan, Pollution to product—a novel two-stage single-pot fermentative production of 1,3-Propanediol, Bioenergy Res. 16 (3) (Sep. 2023) 1528–1536, https://doi.org/10.1007/s12155-023-10570-1.

[7] T. Boonsongsawat, et al., Solvent extraction of biologically derived 1,3-propanediol with ethyl acetate and ethanol cosolvent, Sep. Sci. Technol. 45 (4) (Jan. 2010) 541–547, https://doi.org/10.1080/01496390903526303.

[8] A. Müller, A. Górak, Extraction of 1,3-propanediol from aqueous solutions using different ionic liquid-based aqueous two-phase systems, Separation Purification Technol. (Sep. 2012) 130–136, https://doi.org/10.1016/j.seppur.2012.02.039.

[9] W.B. Sui, Y.Q. Sun, X.L. Wang, Z.L. Xiu, Synergistic extraction of 1,3-Propanediol from fermentation broths using multialcohol Extractants, ACS Sustain Chem Eng 10 (36) (Sep. 2022) 11891–11901, https://doi.org/10.1021/acssuschemeng.2c02853.

[10] Z. Li, L. Yan, J. Zhou, X. Wang, Y. Sun, Z.L. Xiu, Two-step salting-out extraction of 1,3-propanediol, butyric acid and acetic acid from fermentation broths, Sep. Purif. Technol. 209 (Jan. 2019) 246–253, https://doi.org/10.1016/J.SEPPUR.2018.07.021.

[11] V. Liberato, F.S. Buarque, T.F. Ferreira, B.D. Ribeiro, M.A.Z. Coelho, Salting-out as a fast and simple method for first-step separation of biobased 1,3-Propanediol, J. Chem. Eng. Data (2023), https://doi.org/10.1021/ACS.JCED.3C00710/ASSET/IMAGES/LARGE/JE3C00710_0003.JPEG.

[12] X.H. Liu, M. Rebroš, I. Dolejš, A.C. Marr, Designing ionic liquids for the extraction of alcohols from fermentation broth: Phosphonium alkanesulfonates, solvents for diol extraction, ACS Sustain. Chem. Eng. 5 (9) (Sep. 2017) 8260–8268, https://doi.org/10.1021/acssuschemeng.7b01934.

[13] G. Kaur, H. Kumar, M. Singla, Diverse applications of ionic liquids: a comprehensive review, J. Mol. Liq. 351 (Apr. 2022) 118556, https://doi.org/10.1016/J.MOLLIQ.2022.118556.

[14] W.Y. Lee, K.S. Kim, J.K. You, Y.K. Hong, Effect of cations in ionic liquids on the extraction characteristics of 1,3-Propanediol by ionic liquid-based aqueous biphasic systems, ACS Sustain Chem Eng 4 (2) (Feb. 2016) 572–576, https://doi.org/10.1021/acssuschemeng.5b00893.

[15] Y.V. Rajyaguru, J.H. Patil, R. Kusanur, Ionic liquids, an asset in extraction techniques–a comprehensive review, Rev. Adv. Chem. 2022 12:2 12 (2) (Sep. 2022) 107–122, https://doi.org/10.1134/S2634827622020040.

[16] P. Wang, D. Xu, P. Yan, J. Gao, L. Zhang, Y. Wang, Separation of azeotrope (ethanol and ethyl methyl carbonate) by different imidazolium-based ionic liquids: ionic liquids interaction analysis and phase equilibrium measurements, J. Mol. Liq. 261 (Aug. 2018) 89–95, https://doi.org/10.1016/J.MOLLIQ.2018.04.001.

[17] R. Akshat, A. Bharti, P. Padmanabhan, Atomistic molecular dynamics simulation and COSMO-SAC approach for enhanced 1,3-propanediol extraction with imidazolium-based ionic liquids, J. Mol. Model. 30 (6) (Jun. 2024) 164, https://doi.org/10.1007/s00894-024-05964-7.

[18] R. Akshat, A. Bharti, P. Padmanabhan, Molecular dynamics simulation study on the selective extraction of 1,3 propanediol from fermentation broth using imidazolium-based hydrophobic ionic liquids, Fluid Phase Equilib. 599 (2026) 114530, https://doi.org/10.1016/j.fluid.2025.114530.

[19] Q. Zhang, K. De Oliveira Vigier, S. Royer, F. Jérôme, Deep eutectic solvents: syntheses, properties and applications, Chem. Soc. Rev. 41 (21) (Oct. 2012) 7108–7146, https://doi.org/10.1039/C2CS35178A.

[20] D.J.G.P. Van Osch, L.F. Zubeir, A. Van Den Bruinhorst, M.A.A. Rocha, M.C. Kroon, Hydrophobic deep eutectic solvents as water-immiscible extractants, Green Chem. 17 (9) (Sep. 2015) 4518–4521, https://doi.org/10.1039/C5GC01451D.

[21] T. Janković, A.J.J. Straathof, A.A. Kiss, Eco-efficient downstream processing of 1,3-propanediol applicable to various fermentation processes, Process Biochem. 143 (Aug. 2024) 210–224, https://doi.org/10.1016/j.procbio.2024.04.040.

[22] M. Laura, T. Monica, V. Dan-Cristian, The effect of crude glycerol impurities on 1,3-propanediol biosynthesis by *Klebsiella pneumoniae* DSMZ 2026, Renew. Energy 153 (Jun. 2020) 1418–1427, https://doi.org/10.1016/j.renene.2020.02.108.

[23] P. Dhakal, J.K. Shah, Recent advances in molecular simulations of ionic liquid–ionic liquid mixtures, Aug. 01, Elsevier B.V. (2019), https://doi.org/10.1016/j.cogsc.2019.02.009.

[24] R. Akshat, A. Bharti, P. Padmanabhan, 1,3-Propanediol extraction from aqueous phase using hydrophobic eutectic solvents: insights from experiments and molecular dynamics simulations, J. Mol. Liq. 423 (2025) 127026, https://doi.org/10.1016/j.molliq.2025.127026.

[25] M.D. Hanwell, D.E. Curtis, D.C. Lonie, T. Vandermeerschd, E. Zurek, G. R. Hutchison, Avogadro: an advanced semantic chemical editor, visualization, and analysis platform, J. Chem. 4 (1) (Aug. 2012), https://doi.org/10.1186/1758-2946-4-17.

[26] D. Van Der Spoel, E. Lindahl, B. Hess, G. Groenhof, A.E. Mark, H.J.C. Berendsen, GROMACS: fast, flexible, and free, J. Comput. Chem. 26 (16) (Dec. 2005) 1701–1718, https://doi.org/10.1002/JCC.20291.

[27] M. Udier-Blagović, P. Morales De Tirado, S.A. Pearlman, W.L. Jorgensen, Accuracy of free energies of hydration using CM1 and CM3 atomic charges, J. Comput. Chem. 25 (11) (Aug. 2004) 1322–1332, https://doi.org/10.1002/JCC.20059.

[28] L.S. Dodda, J.Z. Vilseck, J. Tirado-Rives, W.L. Jorgensen, 1.14*CM1A-LBCC: localized bond-charge corrected CM1A charges for condensed-phase simulations, J. Phys. Chem. B 121 (15) (Apr. 2017) 3864–3870, https://doi.org/10.1021/acs.jpcb.7b00272.

[29] L.S. Dodda, I.C. De Vaca, J. Tirado-Rives, W.L. Jorgensen, LigParGen web server: an automatic OPLS-AA parameter generator for organic ligands, Nucleic Acids Res. 45 (W1) (Jul. 2017) W331–W336, https://doi.org/10.1093/nar/gkx312.

[30] D. Vassetti, M. Pagliai, P. Procacci, Assessment of GAFF2 and OPLS-AA general force fields in combination with the water models TIP3P, SPCE, and OPC3 for the solvation free energy of Druglike organic molecules, J. Chem. Theory Comput. 15 (3) (Feb. 2019) 1983–1995, https://doi.org/10.1021/ACS.JCTC.8B01039.
[31] B. Hess, H. Bekker, H.J.C. Berendsen, J.G.E.M. Fraaije, LINCS: a linear constraint solver for molecular simulations, J. Comput. Chem. 18 (1997) [Online]. Available: https://api.semanticscholar.org/CorpusID:21320216.
[32] S. Nosé, M.L. Klein, A study of solid and liquid carbon tetrafluoride using the constant pressure molecular dynamics technique, J. Chem. Phys. 78 (11) (Jun. 1983) 6928–6939, https://doi.org/10.1063/1.444641.
[33] W.G. Hoover, Canonical dynamics: equilibrium phase-space distributions, Phys Rev A (Coll Park) 31 (3) (Mar. 1985) 1695, https://doi.org/10.1103/PhysRevA.31.1695.
[34] M. Parrinello, A. Rahman, Polymorphic transitions in single crystals: a new molecular dynamics method, J. Appl. Phys. 52 (12) (Dec. 1981) 7182–7190, https://doi.org/10.1063/1.328693.
[35] M. Brehm, M. Thomas, S. Gehrke, B. Kirchner, TRAVIS—A free analyzer for trajectories from molecular simulation, J. Chem. Phys. 152 (16) (Sep. 2020) 164105, https://doi.org/10.1063/5.0005078/198859.
[36] W. Humphrey, A. Dalke, K. Schulten, VMD – visual molecular dynamics, J. Mol. Graph. 14 (1996) 33–38.
[37] A.D. Wade, A.P. Bhati, S. Wan, P.V. Coveney, Alchemical free energy estimators and molecular dynamics engines: accuracy, precision, and reproducibility, J Chem Theory Comput 18 (6) (Jun. 2022) 3972–3987, https://doi.org/10.1021/acs.jctc.2c00114.
[38] C.H. Bennett, Efficient estimation of free energy differences from Monte Carlo data, J. Comput. Phys. 22 (2) (1976) 245–268, https://doi.org/10.1016/0021-9991(76)90078-4.
[39] A. Luzar, Resolving the hydrogen bond dynamics conundrum, J. Chem. Phys. 113 (23) (Dec. 2000) 10663–10675, https://doi.org/10.1063/1.1320826.
[40] A. Luzar, D. Chandler, Hydrogen-bond kinetics in liquid water, Nat. 1996 379:6560 379 (6560) (Jan. 1996) 55–57, https://doi.org/10.1038/379055a0.
[41] S. Barani Pour, J. Jahanbin Sardroodi, A. Rastkar Ebrahimzadeh, M.S. Avestan, Structural and dynamic properties of eutectic mixtures based on menthol and fatty acids derived from coconut oil: a MD simulation study, Sci. Rep. 12 (1) (2022) 5153, https://doi.org/10.1038/s41598-022-09185-x.